Can One Quantum Universe Run Every Computable Mind?

Gary Abraham Bernstein

Independent Researcher ORCID: https://orcid.org/0009-0009-1761-2867

Abstract

Give every finite program a turn, repeatedly, and each can receive arbitrarily many steps. This mathematical schedule suggests a route to running every computable mind. Can one quantum universe supply it? The paper separates the schedule from the physical execution and from the claim that an execution experiences anything. Unlimited continuation requires the relevant resources; a universal computation does not establish that our universe provides them. Nor does computability include every mathematical structure. The result is a precise target: show which programs the physical model executes, which executions constitute minds, and how their observations are weighted.

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1. The Setup

An ideal computer can run many programs by giving each a turn. With unbounded time and resources, a suitable schedule gives every finite program arbitrarily many steps. The question here is whether a quantum universe can physically realize that schedule, and which of the resulting computations would be experiences.

The path integral formulation was developed by Richard Feynman, with an important precursor in Paul Dirac's work. It calculates a transition amplitude by summing contributions over paths; Feynman's 1948 paper presents its equivalence with ordinary quantum mechanics. A Feynman diagram instead organizes terms in a perturbative expansion. Neither each diagram nor each integration path is automatically a literal particle trajectory or an Everettian world. See Feynman, Reviews of Modern Physics 20, 367 (1948), https://authors.library.caltech.edu/records/9h858-5hv71.

For two alternatives, the probability involves $|a_1 + a_2|^2$, including an interference term, rather than generally $|a_1|^2 + |a_2|^2$. Independent classical alternatives do not have that rule. Taking the amplitude structure as real is a proposed ontological reading; assigning every term its own traversing particle requires an additional argument. Mathematical existence of a path and membership as a constituent history of our universe are distinct claims, even within mathematical monism.

The Mathematical Universe Hypothesis (Tegmark, 1998, 2014) extends ontological commitment further: every consistent mathematical structure exists as a physical universe of its own. The Level IV multiverse includes structures with different fundamental constants, different gauge groups, different dimensions, and different mathematics. Tegmark's argument is, in one form, that any external selection principle separating "real" from "merely possible" mathematical structures would itself be a mathematical structure requiring explanation; the only non-regressive position is no filter at all.

The question this paper raises: is Level III (MWI branching within our universe) already sufficient for everything we would want to call existence? Or does Level IV add something Level III cannot reach?

Some parts of this comparison admit mathematical proofs, including equivalence of formulations and the limits of an effective enumeration. Those implications do not require experimental validation. An a priori argument may also address ontological realization, but it must supply its premises and inference. The demand here is for the missing logical steps, not a new experiment for every metaphysical claim.

2. Why Level III Might Be Sufficient

Three considerations push toward Level III sufficiency.

Fix a description language and a class of complete candidate models. With assigned code weights w(H) proportional to $2^{\wedge}(-\mathrm{L}(\mathrm{H}))$, a genuine difference of ΔL bits gives a prior ratio $2^{\wedge}\Delta\mathrm{L}$. Ten bits gives exactly 1,024. Posterior odds additionally multiply by the evidence likelihood ratio. Summing over all programs producing an output is different from assigning only its shortest program weight. Neither construction automatically defines a normalized measure over every mathematical structure or every observer.

A universal machine can dovetail finite computations of all programs, so every finite stage of each total computable history can eventually be produced. This does not identify all total programs, finish every infinite history or show that our universe executes the enumeration. Physical computational universality and resources must be established separately.

2.3 Arithmetic encoding. Gödel numbering encodes finite expressions and proofs of effectively presented formal systems. It does not make arithmetic prove every encoded truth, implement every encoded system or represent every arbitrary infinite structure by a finite program. A physical machine doing arithmetic demonstrates finite operations, not unlimited cosmological resources. The forest can be described inside a tree without being physically grown there.

3. What Would Survive Under Level III Alone

One wavefunction supplies the states reachable from its initial condition under its dynamics. Consistency with the laws is a weaker requirement than reachability from that condition. Moreover, combining the Standard Model with general relativity into a complete quantum cosmological dynamics is part of the problem being studied, not a supplied Hamiltonian in this paper.

Candidate phenomena include different measurement records, cosmological developments and physical implementations of minds. Each requires support in the specified evolution. Boltzmann fluctuations additionally depend on the cosmology, its state and its dynamics. No arbitrary configuration or complete imagined life is guaranteed by quantum branching alone.

A small addition to a description need not be a dynamically possible fluctuation in the described universe. A comparison with Boltzmann brains requires a specified cosmology and measure, including rates and observer weights. A short verbal addition supplies no probability by itself.

A false memory can represent an extended life without the represented events having occurred. It therefore cannot be counted as the complete series of experiences it describes. A theory that attributes experience to a brief fluctuation must specify the relevant organization and duration.

4. What Would Not Survive

Within a specified universal dynamics, branches share the governing theory. Their effective descriptions can nevertheless differ, for example through different states of symmetry breaking fields. Distinguish a change of effective parameters from replacing the fundamental theory.

Structures requiring different fundamental dynamics are not automatically branches of the same state. Different algebraic structures can also be studied, but an inconsistent claim about fixed standard integers is not made consistent by placing it in a different universe. A different noncommutative ring is a different structure.

These would require either: (a) different effective physics within different inflationary regions, as in the string-landscape solution within Level II; or (b) qualitatively different mathematical structures, as in Level IV proper.

The Schmidhuber/Gödel argument complicates this. If our universe's closure is computationally universal (and any closure containing a Turing machine is), then the dynamics of any other consistent computable structure can be simulated within ours. The question this paper raises is whether such simulation suffices for ontological existence of the simulated structure, or whether the simulated structure has a separate existence in addition.

5. A Further Constraint: Decidability of Consistency

Consistency is a property of a set of axioms or a formal description relative to its logic. An actual mathematical structure satisfies its own truths. The computational question must specify an effective input, such as an index for an enumerator of axioms, rather than assume arbitrary infinite structures have finite encodings.

For effectively axiomatized theories in suitable expressive settings, there is no algorithm deciding consistency in every case. If a contradiction has a finite proof in the chosen proof system, proof search can eventually find it. Failure to find one at a finite stage does not certify consistency. These are precise limitations on methods; they do not imply that every particular consistency question is undecidable.

Universal realization is a further ontological commitment. If every consistent structure is physically realized, consistent selection rules are included. The reverse inference does not follow: the consistency of a filter, including the filter that excludes nothing, does not establish its physical realization without assuming the ensemble being argued for. Mathematical monism motivates the proposal but does not by itself prove that every mathematical possibility is physical.

A fictional sketch can be incomplete without being inconsistent. Some sketches admit consistent completions, others contain incompatible stipulations, and some cases remain undecided. No percentage of fictions in those classes has been established here. General undecidability is not a proof that a specified fictional world cannot be completed.

The general decision problem has limits even though individual cases can be resolved. This distinguishes a missing calculation from the absence of a universal deciding procedure.

Describing a character as afraid does not supply an experiencing implementation. Completing that description would require an account of its relevant organization. Whether the resulting system counts as that fictional character is a further identity question. No conclusion about the prevalence of conscious completions follows from the sketch alone.

6. Fictions as Experiences, Not Universes

The decidability constraint and the consciousness-containing-closure refinement converge on a sharper formulation of the fiction question, with implications for both the Level III/Level IV comparison and the broader mathematical-monist position.

The claim that every fiction is real conflates a description, its possible completions and a physical implementation. Incompleteness alone proves neither

consistency nor inconsistency. The distinctions remain necessary even if universal realization is assumed.

A fictional world need not be physically realized for someone to experience a representation of it. A dream is a process in an experiencing system; its represented content can violate the rules of that system's surroundings. Extending this to standalone minds requires both the universal realization premise and an implementation criterion for conscious dynamics.

Such a mind can represent anything its computational machinery permits. Its representational states can encode physics-violating content, impossible worlds, magic. The mind follows consistent computational rules. The content it represents need not independently exist. A dream of Hogwarts is real computation in a real mind. Hogwarts need not have its own physics. Two such minds, causally connected, can constitute a "universe" of two sentient beings whose shared world is the representational content of their connection.

Under those additional premises, some otherwise unrealized scenarios could occur as represented experiences. This is distinct from proving that every fiction has an experiencing subject.

If universal realization is adopted as a premise, it may include candidate minds with no implementation in our particular cosmology. Whether any such candidate really lies beyond that cosmology requires an account of its organization and the cosmology's reachability. The null filter does not provide a constructive guarantee. The question remains whether one specified dynamics reaches every implementation in a stated class.

The halting problem does not show that some experiences are impossible. A computation can continue indefinitely while producing finite outputs along the way. A perpetual stream needs no final halt to produce its next symbol. Whether a proposed experience has a computable or physically available implementation is a separate question.

The contribution is a separation of the existence of represented content from the implementation of a representing mind. Whether a cosmological wavefunction reaches every relevant implementation remains an open physical question.

7. Caveats and Residuals

Three distinctions prevent a subsumption conclusion from following merely from universality.

Ongoing computation. A finite region cannot finish an infinite sequence, but it can implement finite portions of an ongoing process. In a monotone program model, programs that keep producing output can contribute weight to finite observed prefixes. Nontermination is not automatically zero probability; the probability construction must specify which outputs it counts.

Noncomputable structures. A prior over finite programs does not automatically cover arbitrary noncomputable structures. A computable stochastic law can nevertheless assign probability one to a class of individually noncomputable infinite outcomes. The zero probability of each singleton does not give zero probability to that whole class.

Implementation beyond representation. A mathematical description can specify relations that a particular physical system does not implement. Even among computable candidates, resource limits and unreachable initial conditions can matter. These are substantive obstacles without introducing a nonstructural substance.

None of these distinctions licenses assigning zero total weight to the excluded class. A prior over finite outputs, a measure on infinite sequences and a physical observer measure have different domains. A relation between them has to be supplied.

8. Conclusion

The scheduling construction is exact: at stage s, advance each of the first s programs by one step. Every fixed program receives arbitrarily many steps if the schedule continues. Each finite stage requires only finite resources.

Applying that result to a universe requires a physical account that supplies the continuing execution. Applying it to experience requires a criterion for which computations constitute minds. With those premises, one quantum universe could contain the corresponding computable observers. Comparing it with a larger collection of worlds also requires the observers' relative weights.

References

Everett, H. (1957). "Relative State" Formulation of Quantum Mechanics. *Reviews of Modern Physics* 29, 454.

Feynman, R. P. (1948). Space-Time Approach to Non-Relativistic Quantum Mechanics. *Reviews of Modern Physics* 20, 367.

Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik* 38, 173, 198.

Matiyasevich, Y. (1970). Enumerable sets are Diophantine. *Doklady Akademii Nauk SSSR* 191, 279, 282.

Schmidhuber, J. (1997). A Computer Scientist's View of Life, the Universe, and Everything. In *Foundations of Computer Science: Potential - Theory - Cognition*, ed. C. Freksa, Springer.

Schmidhuber, J. (2000). Algorithmic Theories of Everything. arXiv:quant-ph/0011122.

Solomonoff, R. (1964). A Formal Theory of Inductive Inference. *Information and Control* 7, 1, 22 and 224, 254.

Tegmark, M. (1998). Is "the theory of everything" merely the ultimate ensemble theory? *Annals of Physics* 270, 1.

Tegmark, M. (2014). *Our Mathematical Universe: My Quest for the Ultimate Nature of Reality.* Knopf.

Turing, A. M. (1936). On Computable Numbers, with an Application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society* 42, 230, 265.

Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory According to the Everett Interpretation.* Oxford University Press.